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1 cd "F:\ECON 408\Practice do file"
2
3
4 clear all
5 set seed 1000
6
7 * Task 1:
8 *-----
9
10
11 import excel "USD to BDT.xlsx", firstrow clear
12
13 rename Close Ex_Rate
14 label variable Ex_Rate "Exchange Rate"
15
16 replace Date = dofc(Date)
17 format Date %td
18 tsset Date
19
20
21 * Task 2:
22 *-----
23
24
25 set scheme cleanplots
26
27 tsline Ex_Rate, ///
28         lcolor(navy) lwidth(medthick) ///
29         title("{bf:Daily Exchange Rate Trend (2011-2025)}",
30 size(large)) ///
31         xtitle("Year", size(medlarge)) ///
32         ytitle("Rate (BDT)", size(medlarge)) ///
33         ylabel(, labsize(medium)) ///
34         tlabel(01jan2011 01jan2016 01jan2021 01jan2026,
35 format(%tdcY) labsize(medium)) ///
36         graphregion(color(white)) xsize(8) ysize(4) ///
37         name(tsplot, replace) note("Source: Google Finance")
38
39 graph export Hw_1_tsplot.emf, name(tsplot) replace
```

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36
37
38 * Task 3:
39 *-----
-----
40
41 bcal create bus_cal, from(Date) gen(time) replace
42 format time %tbbus_cal
43 tsset time
44 gen Diff_Ex = D.Ex_Rate
45
46
47 * Task 4:
48 *-----
-----
49
50 * Step 1: Stationarity Check
51
52 tsset Date
53
54 set scheme cleanplots
55
56 tsline Diff_Ex, ///
57         lcolor(navy) lwidth(medthick) ///
58         title("{bf:Daily Diff_Ex Plot (2011-2025)}", size(
59 large)) ///
60         xtitle("Year", size(medlarge)) ///
61         ytitle("Rate (BDT)", size(medlarge)) ///
62         ylabel(, labsize(medium)) ///
63         tlabel(01jan2011 01jan2016 01jan2021 01jan2026,
64 format(%tdcY) labsize(medium)) ///
65         graphregion(color(white)) xsize(8) ysize(4) ///
66         name(Diff_Ex_plot, replace)
67
68 graph export Diff_Ex_plot.emf, name(Diff_Ex_plot) replace
69
70 dfuller Diff_Ex
71 tsset time
72

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71 * Step 2: Empirical ACF and PACF check
72 set scheme s2color
73
74 ac Diff_Ex, name(acf_plot,replace) lags(20)
75 pac Diff_Ex, name(pacf_plot,replace) lags(20)
76 graph combine acf_plot pacf_plot, title("ACF and PACF of
Diff_Ex") xsize(8) ysize(4) name(acf_pacf_plot, replace)
77
78 graph export hw_1_acf_pacf_plot.emf, name(acf_pacf_plot)
replace
79
80 *ARMA(2,3) , ARMA(3,2) , ARMA(3,3) , ARMA(3,4), ARMA(4,3)
81
82
83 qui arima Diff_Ex, ar(1/2) ma(1/3)
84 estat ic
85
86 qui arima Diff_Ex, ar(1/3) ma(1/2)
87 estat ic
88
89 qui arima Diff_Ex, ar(1/3) ma(1/3)
90 estat ic
91
92 qui arima Diff_Ex, ar(1/3) ma(1/4)
93 estat ic
94
95 qui arima Diff_Ex, ar(1/4) ma(1/3)
96 estat ic
97
98
99 * ARMA(3,2) Model has the lowest AIC and BIC
100
101 * Step 3: Estimation
102
103 arima Diff_Ex, ar(1/3) ma(1/2)
104 estat ic
105 estat aroots , name(arootshw1, replace)
106 graph export arootshw1.emf, name(arootshw1) replace
107 predict res , residuals

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```
108
109 ac res, name(acf_res,replace) lags(10)
110 pac res, name(pacf_res,replace) lags(10)
111 graph combine acf_res pacf_res, title("ACF and PACF of
Residuals") xsize(8) ysize(4) name(hw1_res_ac_pac_plot,
replace)
112 graph export hw1_res_ac_pac_plot.emf, name(
hw1_res_ac_pac_plot) replace
113
114
115 wntestq res, lags(10)
116
117
118
```